

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

A UNIVERSE OF LATENT VARIABLE MODELS

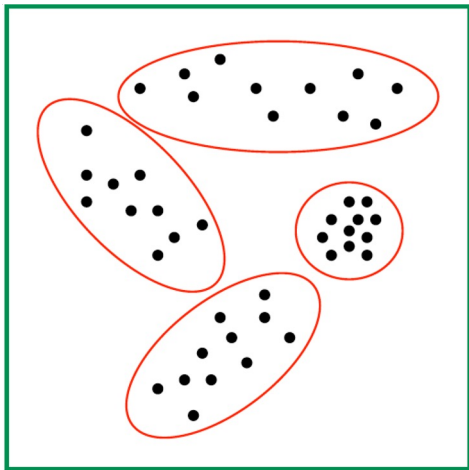
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A model of clustered data: mixture of k Gaussians

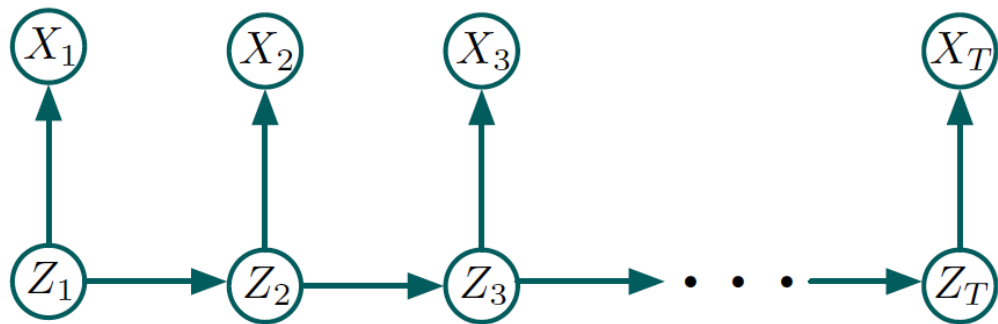
$$\pi_1 N(\mu_1, \Sigma_1) + \pi_2 N(\mu_2, \Sigma_2) + \cdots + \pi_k N(\mu_k, \Sigma_k)$$



- $Z \sim \text{categorical}(\pi_1, \dots, \pi_k)$
- $X \sim N(\mu_z, \Sigma_z)$

Hidden Markov Models

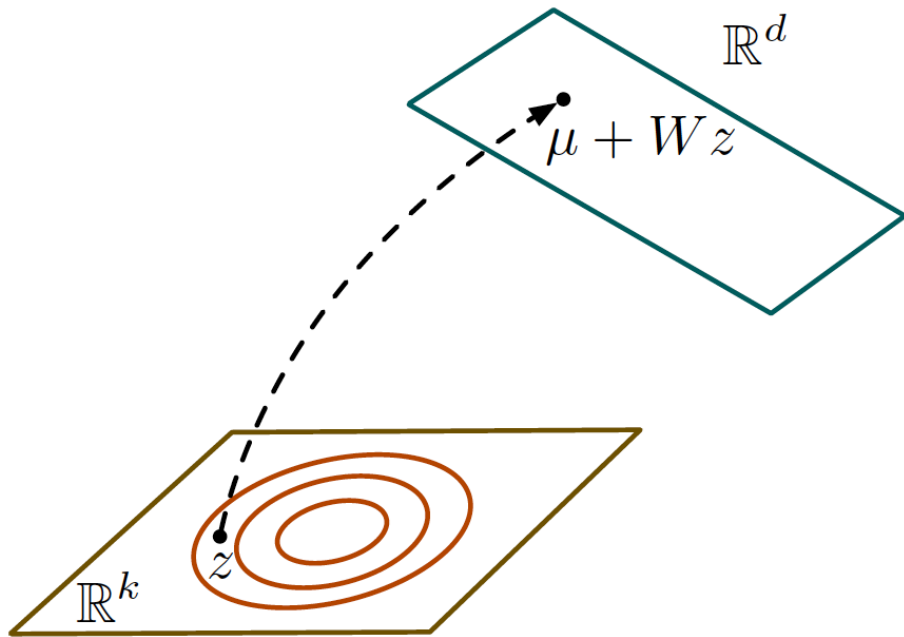
A model of sequence data: HMM with initial distribution π ; state distributions $P_1, \dots, P_k$; transition matrix A .



- $Z_1 \sim \text{categorical}(\pi)$
- For $t = 1, 2, \dots, T$:
 - $X_t \sim P_{Z_t}$
 - $Z_{t+1} \sim \text{categorical}(A_{Z_t})$

Factor Analysis

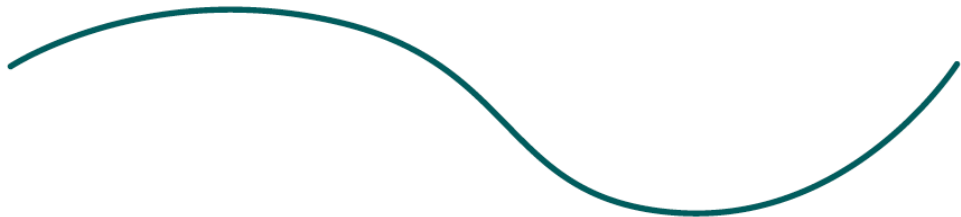
A model of data close to a linear subspace: Factor analysis model with parameters μ, W, Σ .



- $Z \sim N(0, I_k)$
- $X \sim N(\mu + WZ, \Sigma)$

Mixtures of Factor Analyzers

A model of data close to a manifold: Mixture of m factor analyzers, with Parameters $\pi \in \Delta_m; (\mu_1, W_1), \dots, (\mu_m, W_m); \Sigma$.



- $J \sim \text{categorical}(\pi)$
- $Z \sim N(0, I_k)$
- $X \sim N(\mu_J + W_J Z, \Sigma)$

And Many More

- **Topic models:** models of collections of documents
- **Image models:** models of natural images
- **Social network models:** models of influence propagation among interacting people
- **Phylogenetic models:** models of evolutionary trees
- $\vdots$

In each case: a probabilistic model whose underlying graph has a combination of observable and hidden (latent) variables.